

Helix Thready — Troubleshooting

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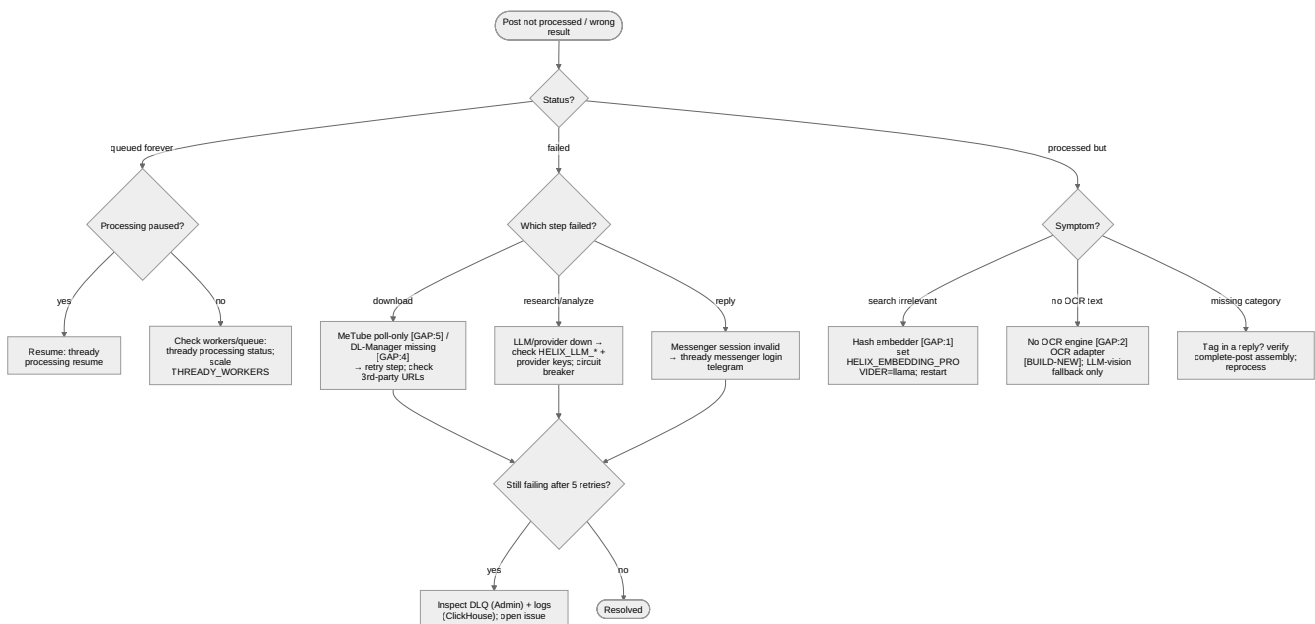
Rev	Date	Author	Change
1	2026-07-21	swarm (user-guides)	Initial symptom → cause → fix guide
2	2026-07-22	swarm (user-guides, Pass 3)	Depth pass: corrected messenger names to VERIFIED Herald <code>HERALD_MTPROTO_*</code> / <code>HERALD_TGRAM_*</code> ; split the triage diagram explanation into multi-paragraph form; added §11 environment-variable & performance troubleshooting matrix and §12 quick symptom index; renumbered Open items to §13.

Symptom-first troubleshooting. Many entries reference **known scaffold traps** from the [gap register](#) — these are not bugs in your setup, they are documented `[BUILD-NEW]` /stub statuses you should recognize. Start with `thready doctor` ([cli-reference.md §5.8](#)).

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1. Post-processing triage (diagram)



Rendered PNG/SVG exported via Docs Chain (\$11.4.65). Source: [diagrams/troubleshoot-post.mmd](#).

Explanation (for readers/models that cannot see the diagram). The triage starts from a post that either never processed or produced a wrong result, and its first and most important branch is on the post's **status**,

because “never ran” and “ran wrong” have entirely different root causes. Reading the status (`thready_post status <id>`) is therefore step zero of every download/processing complaint.

If the post is *queued forever*, the two candidate causes are a **paused pipeline** and a **starved worker pool**. Check pause first (`thready_processing status` ; resume if paused) because it is the cheaper explanation and the more common operator mistake; only if processing is running do you look at queue depth and raise `THREADY_WORKERS`. A queued-forever post is never lost — it is waiting, so the fix is always to restore capacity, never to re-submit.

If the post *failed*, branch on the **step** that failed, because each step has a distinct failure domain. A `download` failure usually traces to the two documented download gaps (MeTube is poll-only `[GAP:5]`, the generic Download Manager doesn’t exist yet `[GAP:4]`) or to a dead 3rd-party URL, and is retried per-step. A `research` / `analyze` failure points at the LLM stack — check `HELIX_LLM_*`, the provider keys, and whether the circuit breaker tripped on a flapping provider. A `reply` failure means the messenger session is invalid, which for Telegram means the `HERALD_MTPROTO_*` session expired and needs a re-login.

If the post *processed but* looks wrong, branch on the **symptom**. Irrelevant **search** is the classic hash-embedder trap `[GAP:1]` — set `HELIX_EMBEDDING_PROVIDER=llama`, re-embed, and restart. Missing **OCR text** is the no-OCR-engine gap `[GAP:2]`, where only the LLM-vision fallback runs until the adapter ships. A **missing category** almost always means the hashtag lived in a *reply* to the root post, so the fix is to confirm complete-post assembly and reprocess.

All failing branches converge on one final check: if the post still fails after the 5-retry ceiling it lands in the **dead-letter queue** for an Admin to inspect alongside the ClickHouse logs; otherwise the per-step retry resolves it and the post reaches `processed`. The DLQ is the single place stuck work accumulates, which is why every branch routes there rather than silently dropping the post.

2. Service won't start

Symptom	Cause	Fix
<code>ABORT: missing required var X</code> (exit 3)	Required env var absent	Add it to <code>.env</code> ; run <code>thready config validate</code> .
<code>ABORT: non-semantic embedder blocked</code>	<code>HELIX_EMBEDDING_PROVIDER=hash</code> <code>[GAP: 1]</code>	Set <code>=llama</code> and point <code>THREADY_EMBEDDING_BASE_URL</code> at a running llama.cpp.
DB connection refused	Postgres/pgvector not up	<code>podman compose ... up -d postgres</code> ; check <code>THREADY_DB_DSN</code> .
<code>pgvector extension not found</code>	Extension not installed	<code>CREATE EXTENSION vector;</code> in the DB.
TLS cert error on boot	Let's Encrypt not issued yet	Check <code>LETS_ENCRYPT_*</code> ; see deployment.

The service **fails loudly and lists** what's missing — read the first error line; it names the key.

3. Can't sign in to a messenger

`[GAP: 3]` **Know the status first.** Telegram works via the `gotd/td` MTPROTO user client (being promoted from the `qahera` harness). **Max is not implemented** — the adapter is `[BUILD-NEW]`.

Symptom	Cause	Fix
messenger 'max' not implemented	Max adapter is [BUILD-NEW] (Herald MAX.md = PLANNED)	Expected in the zero version; use Telegram. Track ATM — Max adapter.
Telegram login code never arrives	Wrong HERALD_MTPROTO_APP_ID / APP_HASH or HERALD_MTPROTO_PHONE	Re-check credentials from my.telegram.org/apps; phone must be E.164.
SESSION_PASSWORD_NEEDED / 2FA loop	Account has cloud 2FA	Set HERALD_MTPROTO_PASSWORD .
Session lost after restart	Session file not persisted	Set HERALD_MTPROTO_SESSION_FILE to a writable, encrypted path (default ~/.config/herald/mtproto.session).
Can't backfill channel history	Using the Bot-API path (HERALD_TGRAM_BOT_TOKEN) — a bot cannot read history	Ensure the MTPROTO user client (HERALD_MTPROTO_*) is active; the bot token is reply-only.
Replies never post but reading works	Bot token/chat unset	Set HERALD_TGRAM_BOT_TOKEN + HERALD_TGRAM_CHAT_ID ; verify THREADY_REPLY_ACCOUNT .

4. Posts never get processed

Symptom	Cause	Fix
All posts <code>queued</code> , none <code>processing</code>	Processing paused	<code>thready processing status ; thready processing resume</code> .
Slow drain under load	Worker pool too small	Raise <code>THREADY_WORKERS</code> (default 32).
A post <code>processing</code> for very long	Research-heavy soft timeout	<code>THREADY_POST_TIMEOUT</code> (default 30 m); long downloads are delegated and shouldn't hold a slot.
System replies being re-processed	Should never happen	Only organic posts are processed by design; if you see this, file it — the reply-exclusion invariant is broken.
Duplicate processing	Should never happen	Single-claim via Postgres lock guarantees exactly-once; report if observed.

5. Semantic search returns irrelevant results

[GAP: 1] **The #1 gotcha.** HelixLLM's default `HashEmbedder` emits non-semantic pseudo-vectors; search built on it returns garbage with only a startup warning.

Fix: 1. `thready doctor embeddings` → if it shows `provider=hash`, that's the cause. 2. Start a real llama.cpp embedding server; set `HELIX_EMBEDDING_PROVIDER=llama` and `THREADY_EMBEDDING_BASE_URL / THREADY_EMBEDDING_MODEL`. 3. Ensure `THREADY_EMBEDDING_DIM` matches the model (the RAG path historically hardcoded 768 — set it explicitly). 4. **Re-embed** existing content: `thready search reindex --all` (embeddings computed on the hash provider must be regenerated). 5. Restart; confirm the portal's "degraded" banner clears.

6. A download never completes

Content	Cause	Status
Video (<code>#Video</code> , <code>#Channel</code>)	MeTube is poll-only <code>[GAP: 5]</code> — no completion webhook	Thready polls <code>GET /api/postprocess/status</code> ; completion notification lags. The download still finishes; the outbound webhook is <code>[BUILD-NEW]</code> .
Torrent (<code>#Torrent</code>)	Boba callback contract bespoke <code>[GAP: 6.4]</code>	Boba has SSE + <code>POST /api/v1/hooks</code> ; if callbacks don't arrive, check <code>THREADY_BOBA_CALLBACK_URL</code> reachability.
Direct file URL (HTTP/FTP/...)	Download Manager doesn't exist yet <code>[GAP: 4]</code>	Direct-URL downloads are unavailable until the <code>[BUILD-NEW]</code> Download Manager ships.
Broken existing asset	Physical link broke	<code>thready asset reheal <id></code> re-downloads.

Use `thready post retry <id> --step download` to re-drive a stuck download step. Persistent failures land in the DLQ.

7. Comic/screenshot text isn't extracted

`[GAP: 2]` VisionEngine has **no OCR engine**. Comic transcription and screenshot text extraction fall back to **LLM-vision** (lower fidelity, non-deterministic, no bounding boxes) until the Tesseract/PaddleOCR adapter `[BUILD-NEW]` lands. `THREADY_OCR_PROVIDER=none` today means “LLM-vision only”. This is expected in the zero version — not a misconfiguration.

8. Auth / token / RBAC errors

Symptom	Cause	Fix
<code>401 unauthorized</code> (exit 4)	Missing/expired token	<code>thready auth login</code> , or set <code>THREADY_TOKEN</code> .
<code>403 forbidden</code> (exit 77)	RBAC denies the action	You lack the role/scope; an Admin must grant it. Not a bug — server enforces.
Tokens rejected across services	HS256 single-service only <code>[GAP: 10]</code>	Move to <code>THREADY_JWT_SIGNING_ALG=RS256</code> with a shared public key.
MFA loop	TOTP not enrolled (Admin)	Complete TOTP enrolment; admin tiers are forced.
API key works but too broad	Over-scoped key	Mint a narrower key: <code>thready auth token --scopes read:search</code> .

9. Configuration changes not taking effect

- **Runtime-editable settings** (directories, poll frequency, branding, retention) go client → REST → System and apply immediately, emitting `config.changed`. If not, check RBAC and the audit log.
- **Secrets are NOT hot-reloaded.** Rotating `THREADY_JWT_SECRET`, provider keys, DB DSN, etc. requires a **service restart**. This is deliberate (secrets are read once at boot).
- **.env precedence:** a value exported in your shell **overrides** the `.env` file. If a change to `.env` “does nothing”, check for a stale `export` in `~/.bashrc` / `~/.zshrc`.

10. Where the logs are

- **Structured logs** — logrus (JSON in prod) shipped to **ClickHouse** via `observability` (`THREADY_CLICKHOUSE_DSN`). Query with the observability tooling; **not** ELK/Loki.
- **Traces** — OpenTelemetry (`OTEL_EXPORTER_OTLP_ENDPOINT`) → Jaeger/Zipkin.
- **Metrics** — Prometheus at `THREADY_METRICS_ADDR` → Grafana.
- **Audit** — `thready audit tail` / `audit query` (append-only).
- **Container logs** — `podman logs <container>` per stack.
- **Secrets are always redacted** in every sink.

11. Environment-variable & performance troubleshooting matrix

Symptoms that trace to a specific misconfigured or under-tuned variable. This complements the [configuration reference](#) — here you start from the *symptom*, there you look up the *variable*.

Symptom	Likely variable	Cause	Fix
Search relevance is garbage	<code>HELIX_EMBEDDING_PROVIDER</code>	Set to <code>hash</code> (or defaulted) <code>[GAP: 1]</code>	Set <code>=llama</code> ; point <code>THREADY_EMBEDDING_BASSE_URL</code> at a live llama.cpp; reindex.
Search returns nothing / dimension error	<code>THREADY_EMBEDDING_DIM</code>	Mismatch with the model's true dim (RAG path historically hardcoded 768)	Set to the model's real dimension (e.g. <code>1024</code>); re-embed.
Search slow (> 500 ms)	<code>THREADY_VECTOR_INDEX</code>	Sequential scan (no ANN index)	Set <code>=hnsw</code> ; ensure the index is built on the embedding column.
Whole system slow under load	<code>THREADY_WORKERS</code> / <code>THREADY_DB_MAX_OPEN_CONNS</code>	Worker or connection-pool starvation	Raise workers; raise pool to match; watch Prometheus queue depth.
Downloads pile up	<code>THREADY_DOWNLOAD_CONCURRENCY</code>	Too few parallel download jobs	Raise it; confirm the 3rd-party service (Boba/ MeTube) is reachable.
LLM steps time out	<code>THREADY_POST_TIMEOUT</code> / <code>HELIX_LLM_BASE_URL</code>	Model host unreachable or budget too tight for research-heavy posts	Verify the llama.cpp host; raise the soft budget; check the circuit breaker.
	<code>THREADY_JWT_SIGNING_ALG</code>		Move to <code>RS256</code> with

Symptom	Likely variable	Cause	Fix
Tokens rejected across services		HS256 is single-service only <code>[GAP: 10]</code>	a shared public key.
Vision de-dupes but never OCRs text	<code>HELIX_VISION_OPENCV_ENABLED</code> vs OCR	OpenCV $\neq$ OCR <code>[GAP: 2]</code>	Expected; OCR adapter is <code>[BUILD-NEW]</code> . OpenCV only finds near-duplicates.
Assets 404 via a direct path	<code>THREADY_ASSET_SERVICE_URL</code>	Client used a raw path instead of resolving through the Asset Service	Always resolve links via the Asset Service; never store direct file paths.
MinIO signed URLs expire mid-download	<code>THREADY_STORAGE_SIGNED_URL_TTL</code>	TTL too short for large assets	Raise it; retry <code>thready asset get</code> .
Config edit “does nothing”	<i>(shell export)</i>	A <code>~/ .bashrc / ~/ .zshrc</code> <code>export</code> overrides <code>.env</code>	Unset the stale export, or change the value there.
Secret rotation not applied	<i>(any secret)</i>	Secrets read once at boot — not hot-reloaded	Restart the service after rotating.

12. Quick symptom index

Jump straight to the section for a one-line symptom:

You see...	Go to
ABORT: missing required var on start	\$2
ABORT: non-semantic embedder blocked	\$2 / \$5
messenger 'max' not implemented	\$3
Login code never arrives	\$3
Posts stuck queued	\$4
Search results irrelevant	\$5
Download never finishes	\$6
Comic/screenshot text missing	\$7
401 / 403 (exit 4/77)	\$8
Variable-specific slowness/errors	\$11

13. Open items

- [OPEN: trb-1] `thready search reindex` depends on the semantic-search service [GAP register \$11] ; confirm the exact reindex command once implemented.
- [OPEN: trb-2] DLQ inspection UX (portal/CLI surface for stuck posts) to be finalized with the BackgroundTasks integration. Tracked: **ATM — DLQ inspection surface**.
- [OPEN: trb-3] Several fixes here reference [BUILD-NEW] services (Max, Download Manager, OCR, MeTube webhook); their operator-facing error messages finalize when those ship.

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